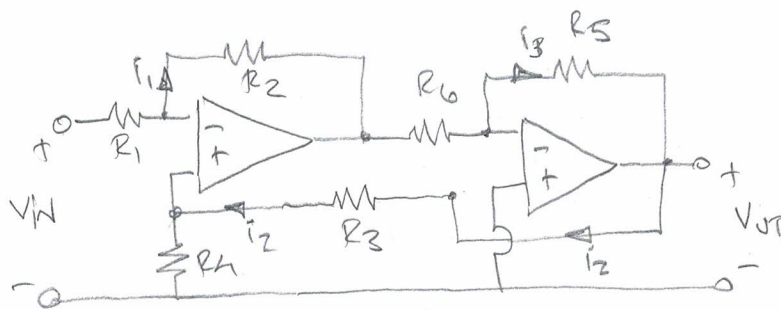


8.35)



ALLA RESISTANSER ÄR LIKA OCH $V_{IN} = 4V$

Lösning: KIRCHHOFF

$$-V_{IN} + R_1 i_1 + R_2 i_1 + R_6 i_3 = 0 \quad (1)$$

$$-V_{OUT} - R_5 i_3 = 0 \quad (2)$$

$$-V_{OUT} + R_3 i_2 + R_4 i_2 = 0 \quad (3)$$

SAKT

$$-V_{IN} + R_1 i_1 + R_4 i_2 = 0 \quad (4)$$

(2) & (4) ger (ALLA R LIKA) (SÄTT ALLA RESISTANSER = R)

$$i_1 = \frac{V_{IN} + V_{OUT}}{2R} \quad (5)$$

$$(3) \text{ ger } i_2 = \frac{V_{OUT}}{2R} \quad (6)$$

(5) & (6) i (4) ger

$$-V_{IN} + R \frac{V_{IN} + V_{OUT}}{2R} + R \frac{V_{OUT}}{2R} = 0$$

$$\underline{V_{OUT} = \frac{1}{2} V_{IN}}$$

SVAR: OM ALLA RESISTANSER
LIKA FÅS

$$V_{OUT} = \frac{1}{2} V_{IN}$$